Automata Theory Exam

Question 1: CFG to CNF Conversion

Convert the given Context-Free Grammar (CFG) into Chomsky Normal Form (CNF).

- **Note:** The given grammar contains:
  - One non-terminal with recursion (must be removed).
  - Two uncalled transitions (unit productions).
  - One Null production (must be handled).

Question 2: PDA Construction

Construct a Pushdown Automaton (PDA) for the Context-Free Grammar (CFG) provided.

Question 3: GNF Conversion

Convert the given CFG into Greibach Normal Form (GNF).

- **Note:** The grammar includes productions like S → AB. The conversion involves removing left recursion, which may result in a large number of terms (approx. 24-30 terms).

Question 4: 2-PDA Construction

Construct a 2-Stack PDA (2-PDA) for the language L = {ww | w ∈ {a, b}*}.

- **Logic:**
  1. Non-deterministically push the first half of the string onto Stack 1.
  2. At the start of the second half, pop from Stack 1 and push to Stack 2 (to reverse the order).
  3. Match the remainder of the input by popping from Stack 2.

Question 5: Turing Machine Design

Design a standard Turing Machine for the language L = {ww}.

- **Logic:**
  1. Find the midpoint of the string.

2. Convert the left half into a normalized format.
3. Map/match the characters of the left half (a, b) to the right half (x, y) to verify the repeat structure.

Question 6: Computable Functions (Square)

Design a Turing Machine that computes the function $f(x) = x^2$ (calculating the square of a number given in unary).

- **Logic:** Copy the string and perform multiplication.

Question 7: Computable Functions (Modulo)

Design a Turing Machine that computes the modulo operation.

- **Logic:** Repeatedly subtract the divisor from the dividend.